

Finite elements

$$u_h(x) = \sum_i c_i \lambda_i(x)$$



basis fixed,
coefficients c_i solved for

d incidence, ± 1 :

the topology

★ mass matrices:

metric, material law



Learn the basis ✗

$$u_\theta(x) = \sum_i c_i \lambda_i^\theta(x)$$



the spaces themselves
become learned objects

$$d \Lambda_h^k \not\subseteq \Lambda_h^{k+1}$$

nothing re-imposes it, and the
bounded commuting projection

has no known analogue



Learn the ★ ✓

$$u_h(x) = \sum_i c_i \lambda_i(x)$$



basis and d untouched;
the network is in the

constitutive law

$\ker d / \operatorname{im} d$ is about d alone
 $\Rightarrow$ the cohomology survives
whatever θ turns out to be